

AVANTEL SINGLE BOARD COMPUTER (SBC)

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1 SINGLE BOARD COMPUTER

a Description

The Intel Atom X6413E is a low-power quad-core processor designed for embedded and industrial applications. It features a 1.5 GHz base frequency, with a boost frequency of up to 3.0 GHz, and integrates Intel UHD Graphics. With a 9W TDP, it offers a balance of performance and power efficiency for applications like industrial automation, control systems, and point-of-sale terminals.

b Block Diagram

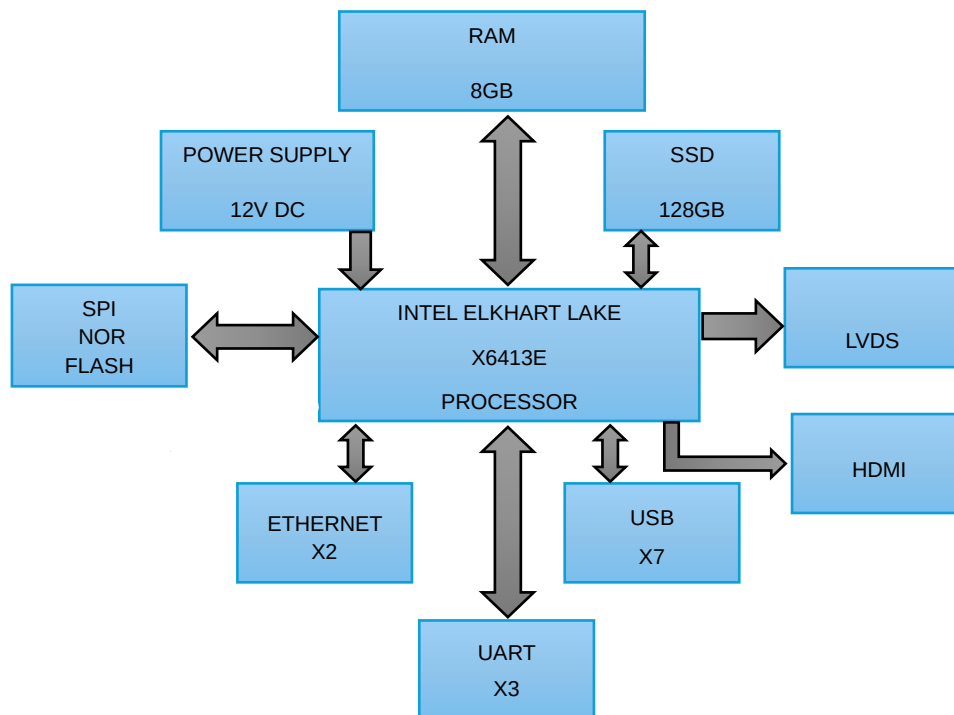


Figure 1: Block Diagram of Single Board Computer (SBC)

c Image

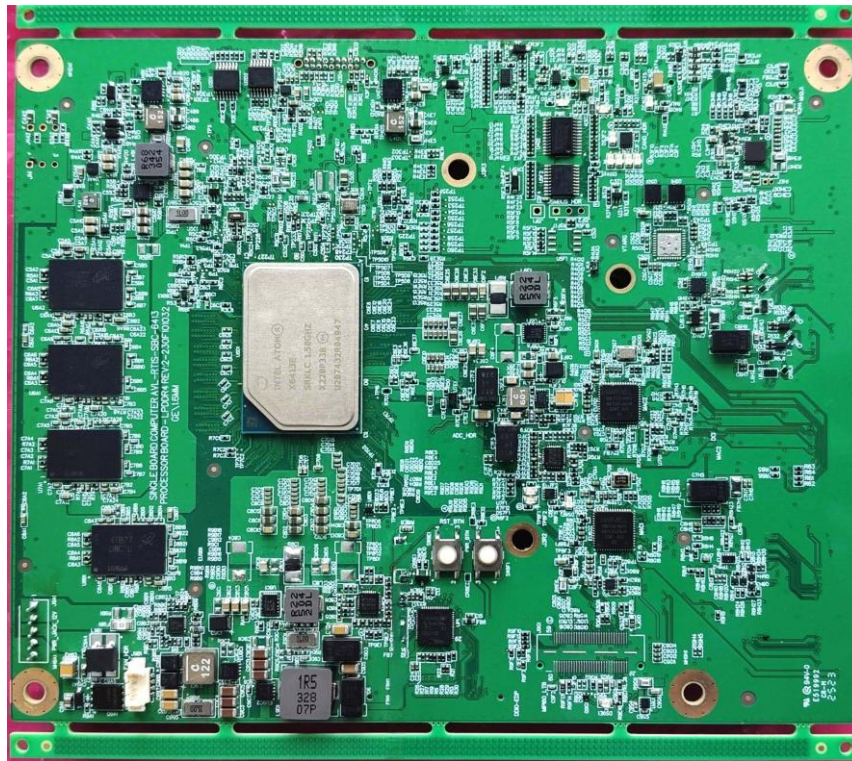


Figure 2: Image of Single Board Computer (SBC)

d Specifications

Table 1: Specifications for SBC

S No.	Parameter	Specifications
1	Processor	Intel Atom® x6413E Processor
2	Core	Quad Core
3	Clock Speed	2.00 GHz
4	TDP (Thermal Design Power)	9.5W
5	Memory	8 GB
6	Storage	128 GB SSD
7	Memory Type	LPDDR4 On Board
8	LVDS	LVDS 18/24bit, Single/Dual Channel Max. resolution up to 1920x1200 @ 60Hz
9	USB Interface	7 x USB
10	HDMI Interface	HDMI 2.0b @ 5.94 Gbps 4K60x3 HDR
11	SSD(M.2)	M.2 slot B-Key 2242
12	Gigabit Ethernet	2 x 1GbE
13	UART	3
14	Processor Series	Elkhart Lake
15	Operating Temperature	-40°C to 85°C
16	Power Supply	12V DC

2 7" LCD DISPLAY

a Description

NJ070NA-23A is model a color active-matrix thin film transistor (TFT) liquid crystal display (LCD) that uses amorphous silicon TFT as a switching device. This model is composed of a TFT LCD panel, a driving circuit and a back light system. This TFT LCD has a 7.0 (16:9) inch diagonally measured active display area with WSVGA (1024 horizontal by 600 vertical pixel array) resolution. The TFT LCD panel used for this module is adapted for a low reflection and higher color type.

b Image



Figure 3: Image of 7" LCD Display

c Specifications

Table 2: Specifications for 7" LCD Display

S No.	Parameter	Specification
1	LCD size	7.0 inch (Diagonal)
2	Pixel size	0.1506(W) × 0.1432(H) mm
3	Active area	154.2144(W) × 85.92(H) mm
4	Module size	164.9(W) × 100(H) × 4.8(D) mm
5	Driver element	a-Si TFT active matrix
6	Weight	140g (Typ.)
7	Display mode	Normally White, Transmissive
8	Resolution	1024 × 3(RGB) × 600
9	Surface treatment	Hard Coating
10	Colour arrangement	RGB-stripe
11	Interface	LVDS
12	View direction (Gray Inversion)	6 O' Clock
13	Backlight power consumption	2.62 W (Typ.)
14	Panel power consumption	0.4 W (Typ.)
15	Touch Screen Type	Capacitive touch
16	Temperature	-40 to +85deg C

3 INNODISK M.2 SATA SSD

a Description

Part Number: DEM24-A28IC1KWADF

Innodisk M.2 (S42) 3TEB provides high-capacity flash memory Solid State Drive (SSD) that electrically complies with Serial ATA (SATA) standard. It supports SATA III standard (6.0GHz) with high performance.

b Image

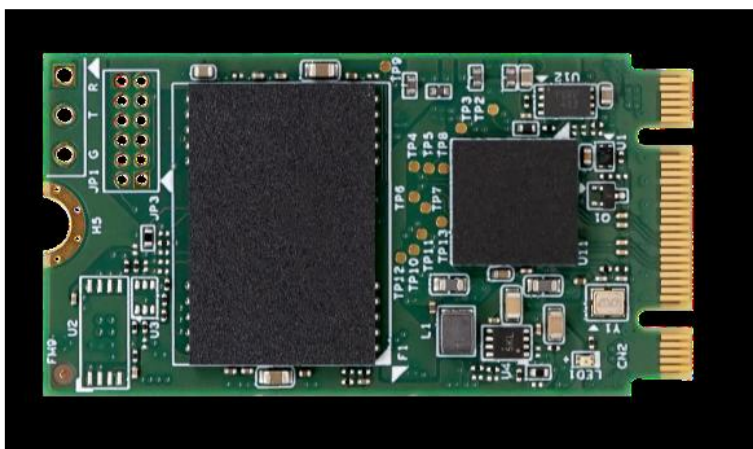


Figure 4: INNODISK M.2 SATA SSD

c Specifications

Table 3: Specifications for INNODISK M.2 SATA SSD

S No.	Parameter	Specification
1	Model Name	M.2 (S42) 3TEB
2	Flash Type	3D TLC
3	Interface	SATA III 6.0 Gb/s
4	Form Factor	M.2 2242 B+M Key
5	Capacity	128GB
6	Sequential R / W (MB/sec, max.)	550 / 480
7	Storage Temperature	-40°C ~ 85°C
8	Max. Power Consumption	1.2W
9	Dimension (W x L x H/mm)	22.0 x 42.0 x 3.5